exp9.py - E:/machine_learning_lab_exp/exp9.py (3.10.11)

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```python
import numpy as np
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import PolynomialFeatures

# Input data (non-linear relationship)
X = np.array([[1], [2], [3], [4]])
y = [1, 4, 9, 16]

# ---------- Linear Regression ----------
linear_model = LinearRegression()
linear_model.fit(X, y)
linear_pred = linear_model.predict([[5]])

# ---------- Polynomial Regression ----------
poly = PolynomialFeatures(degree=2)
X_poly = poly.fit_transform(X)

poly_model = LinearRegression()
poly_model.fit(X_poly, y)
poly_pred = poly_model.predict(poly.transform([[5]]))

print("Input Data:", X.flatten())
print("Actual Output:", y)
print("Linear Regression Prediction for 5:", linear_pred)
print("Polynomial Regression Prediction for 5:", poly_pred)
```

IDLE Shell 3.10.11

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```
Python 3.10.11 (tags/v3.10.11:7d4cc5a, Apr  5 2023, 00:38:17) [MSC v.1929 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
================= RESTART: E:/machine_learning_lab_exp/exp9.py =================
Input Data: [1 2 3 4]
Actual Output: [1, 4, 9, 16]
Linear Regression Prediction for 5: [20.]
Polynomial Regression Prediction for 5: [25.]
>>>
```

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